

Sprint Velocity Estimation Guide

How to estimate sprint velocity for better, data-driven Agile planning

What is Sprint Velocity?

By looking at the amount of work your team completed in previous sprints, you can reliably estimate how much work they can accomplish in future sprints. In Agile development, this key metric and prediction is known as *sprint velocity*.

STEP-BY-STEP VELOCITY ESTIMATION

Step 1: Count completed User Story Points in each sprint

At the end of each sprint, sum up the total number of story points the team successfully completed. Consider the following historic baseline tracking sample:

Sprint	Committed Stories	Points / Story	Total Committed Points	Completed Stories	Completed Points
Sprint 1	5 User Stories	8 Points	40 Story Points	3 Stories	24 Points
Sprint 2	7 User Stories (incl. 2 rolled over)	8 Points	56 Story Points	4 Stories	32 Points
Sprint 3	7 User Stories (incl. 3 rolled over)	8 Points	56 Story Points	5 Stories	40 Points

Step 2: Calculate the average of completed story points

Add up the total number of story points completed across all baseline sprints, then divide by the total number of sprints observed to establish your velocity baseline.

Total Completed Points: $24 + 32 + 40 = 96$ Story Points

Average Sprint Velocity: $96 \div 3 \text{ Sprints} = 32$ Story Points / Sprint

Predictive Planning Application: You can now base the capacity of future sprints on an average of **32 story points**. For instance, if you have **160 story points remaining** in the project backlog, you can systematically assume your team will need approximately **five additional sprints** to complete the project scope.

TRACKING PROGRESS WITH VISUALS

Agile frameworks leverage visual metrics to establish clarity and act as informative clear-box indicators:

Velocity Chart

A visual representation of project progress over time. Its main purpose is to help project managers reliably estimate future team performance.

Velocity Burndown

A graphical "information radiator" that explicitly maps out and visualizes the remaining work planned to be completed within an active sprint window.

Kanban Board

Lists team tasks in columns via structural cards or virtual sticky notes to help you transparently visualize, manage, and track moving work items.

HOW TO STABILIZE AND IMPROVE VELOCITY

Operational Strategies for Consistency

- **Write clean requirements:** Construct user stories that are exceptionally clear, simple to read, and easy to understand for all development disciplines.
- **Maintain stability:** Keep team core membership and overall resource sizing as consistent as possible across development windows.
- **Proactively secure help:** If your delivery team is short on specific technical skills to execute tasks, immediately recruit target inside or outside expert help.

Sprint velocity is not static. Continually reviewing past sprints to see what was accomplished goes a long way toward accurately predicting what your team can confidently deliver in future iterations.